

Magnetism (answer)

9-1

Consider a section of the ribbon of thickness dx located a distance x away from point P . The current it carries is $di = i dx/w$, and its contribution to B_P is

$$dB_P = \frac{\mu_0 di}{2\pi x} = \frac{\mu_0 i dx}{2\pi x w}.$$

Thus,

$$B_P = \int dB_P = \frac{\mu_0 i}{2\pi w} \int_d^{d+w} \frac{dx}{x} = \frac{\mu_0 i}{2\pi w} \ln\left(1 + \frac{w}{d}\right)$$

and $\vec{B}_P$ points **upward**.

9-2

To find the magnitude of the field, use superposition to obtain the result:

$$B = \frac{\mu_0 i \pi}{4\pi a} + \frac{\mu_0 i \pi}{4\pi b} = \frac{\mu_0 i}{4} \left(\frac{1}{a} + \frac{1}{b} \right)$$

(b) By the right-hand rule, $\vec{B}$ points into the paper at P

(c) The enclosed area is $A = (\pi a^2 + \pi b^2)/2$ which means the magnetic dipole moment has magnitude

$$|\vec{\mu}| = \frac{\pi i}{2} (a^2 + b^2).$$

(d) The direction of $\vec{\mu}$ is the same as the $\vec{B}$ found in part (a): **into the paper**.

9-3

(a) The magnetic field at a point within the hole is the sum of the fields due to two current distributions. The first is that of the solid cylinder obtained by filling the hole and has a current density that is the same as that in the original cylinder (with the hole). The second is the solid cylinder that fills the hole. It has a current density with the same magnitude as that of the original cylinder but is in the **opposite** direction. If these two situations are superposed the total current in the region of the hole is zero. Now, a solid cylinder carrying current i which is uniformly distributed over a cross section, produces a magnetic field with magnitude

$$B = \frac{\mu_0 i r}{2\pi R^2}$$

at a distance r from its axis, inside the cylinder. Here R is the radius of the cylinder. For the cylinder of this problem the current density is

$$J = \frac{i}{A} = \frac{i}{\pi(a^2 - b^2)},$$

where $A = \pi(a^2 - b^2)$ is the cross-sectional area of the cylinder with the hole. The current in the cylinder without the hole is

$$I_1 = JA = \pi J a^2 = \frac{i a^2}{a^2 - b^2}$$

and the magnetic field it produces at a point inside, a distance r_1 from its axis, has magnitude

$$B_1 = \frac{\mu_0 I_1 r_1}{2\pi a^2} = \frac{\mu_0 i r_1 a^2}{2\pi a^2 (a^2 - b^2)} = \frac{\mu_0 i r_1}{2\pi (a^2 - b^2)}.$$

The current in the cylinder that fills the hole is

$$I_2 = \pi J b^2 = \frac{i b^2}{a^2 - b^2}$$

and the field it produces at a point inside, a distance r_2 from its axis, has magnitude

$$B_2 = \frac{\mu_0 I_2 r_2}{2\pi b^2} = \frac{\mu_0 i r_2 b^2}{2\pi b^2 (a^2 - b^2)} = \frac{\mu_0 i r_2}{2\pi (a^2 - b^2)}.$$

At the center of the hole, this field is zero and the field there is exactly the same as it would be if the hole were filled. Place $r_1 = d$ in the expression for B_1 and obtain

$$B = \frac{\mu_0 i d}{2\pi (a^2 - b^2)}$$

for the field at the center of the hole. The field points **upward** in the diagram if the current is out of the page.

(b) If $b = 0$ the formula for the field becomes

$$B = \frac{\mu_0 i d}{2\pi a^2}.$$

This correctly gives the field of a solid cylinder carrying a uniform current i , at a point inside the cylinder a distance d from the axis. If $d = 0$ the formula gives $B = 0$. This is correct for the field on the axis of a cylindrical shell carrying a uniform current.

9-4

(a) Our sign choice is such that i is positive in the smaller cylinder and negative in the larger one, for $r < c$.

$$B = \frac{\mu_0 i r}{2\pi c^2}, \quad r \leq c.$$

(b) in the region between the conductors.

$$B = \frac{\mu_0 i}{2\pi r}, \quad c \leq r \leq b.$$

(c) Within the larger conductor we have a superposition of the field due to the current in the inner conductor plus the field due to the (negative) current in that part of the outer conductor at radius less than r . The result is

$$B = \frac{\mu_0 i}{2\pi r} - \frac{\mu_0 i}{2\pi r} \left(\frac{r^2 - b^2}{a^2 - b^2} \right), \quad b < r \leq a.$$

If desired, this expression can be simplified to read

$$B = \frac{\mu_0 i}{2\pi r} \left(\frac{a^2 - r^2}{a^2 - b^2} \right).$$

(d) Outside the coaxial cable, the net current enclosed is zero. So $B = 0$ for $r \geq a$.

10-1

(a) We seek the electrostatic field established by the separation of charges (brought on by the magnetic force). We define the magnitude of the electric field as

$$|\vec{E}| = v|\vec{B}| = (20.0 \text{ m/s})(0.030 \text{ T}) = 0.600 \text{ V/m} .$$

Its direction is opposite to that defined by $\vec{v} \times \vec{B}$. In summary,

$$\vec{E} = -(0.600 \text{ V/m})\hat{k}$$

which insures that $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ vanishes.

$$(b) \quad V = Ed = (0.600 \text{ V/m})(2.00 \text{ m}) = 1.20 \text{ V} .$$

10-2

We note that $\vec{B}$ must be along the x axis because when the velocity is along that axis there is no induced voltage.

$$d = \frac{V}{E} = \frac{V}{vB}$$

where one must interpret the symbols carefully to ensure that $\vec{d}$, $\vec{v}$ and $\vec{B}$ are mutually perpendicular. Thus, when the velocity is parallel to the y axis the absolute value of the voltage (which is considered in the same “direction” as $\vec{d}$) is 0.012 V, and

$$d = d_z = \frac{0.012 \text{ V}}{(3.0 \text{ m/s})(0.020 \text{ T})} = 0.20 \text{ m} .$$

On the other hand, when the velocity is parallel to the z -axis the absolute value of the appropriate voltage is 0.018 V, and

$$d = d_y = \frac{0.018 \text{ V}}{(3.0 \text{ m/s})(0.020 \text{ T})} = 0.30 \text{ m} .$$

Thus, our answers are

(a) $d_x = 25 \text{ cm}$ (which we arrive at “by elimination” – since we already have figured out d_y and d_z),

(b) $d_y = 30 \text{ cm}$, and

(c) $d_z = 20 \text{ cm}$.

10-3

We can obtain the torque vector

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$|\vec{\mu}| = NI A = (20)(0.10 \text{ A})(0.0050 \text{ m}^2)$$

$$\begin{aligned} \tau &= (NI A)(B) \sin\left(\frac{\pi}{2} - \theta\right) = NIAB \cos \theta = (20)(0.10 \text{ A})(0.10 \text{ m})(0.050 \text{ m})(0.50 \text{ T}) \cos 30^\circ \\ &= 0.0043 \text{ N} \cdot \text{m} . \end{aligned}$$

The direction of the torque is $-y$. In unit-vector notation, the torque is

$$\vec{\tau} = (-4.3 \times 10^{-3} \text{ N} \cdot \text{m}) \hat{j}$$

10-4

Consider an infinitesimal segment of the loop, of length ds . The magnetic field is perpendicular to the segment, so the magnetic force on it has magnitude $dF = iB ds$. The horizontal component of the force has magnitude

$$dF_h = (iB \cos \theta) ds$$

and points inward toward the center of the loop. The vertical component has magnitude

$$dF_y = (iB \sin \theta) ds$$

and points upward. Now, we sum the forces on all the segments of the loop. The horizontal component of the total force vanishes, since each segment of wire can be paired with another, diametrically opposite, segment. The horizontal components of these forces are both toward the center of the loop and thus in opposite directions. The vertical component of the total force is

$$F_v = iB \sin \theta \int ds = 2\pi a i B \sin \theta$$

10-5

(a) we have

$$\vec{\mu} = NI A \vec{n} = (NI A)(-\hat{j}) = -(0.0240 \text{ A} \cdot \text{m}^2) \hat{j}$$

$$U = -\vec{\mu} \cdot \vec{B} = -(-0.0240 \text{ A} \cdot \text{m}^2)(-3.00 \times 10^{-3} \text{ T}) = -7.20 \times 10^{-5} \text{ J} .$$

(b) $\vec{\tau} = \vec{\mu} \times \vec{B} = (-0.0240 \hat{j}) \times (2.00 \times 10^{-3} \hat{i}) + (-0.0240 \hat{j}) \times (-4.00 \times 10^{-3} \hat{k})$

$$= (4.80 \times 10^{-5} \hat{k} + 9.60 \times 10^{-5} \hat{i}) \text{ N}\cdot\text{m}.$$

11-1

(a) In the region of the smaller loop the magnetic field produced by the larger loop may be taken to be uniform and equal to its value at the center of the smaller loop, on the axis. With $z = x$ (taken to be much greater than R), we have

$$\vec{B} = \frac{\mu_0 i R^2}{2x^3} \hat{i}$$

where the $+x$ direction is upward. The magnetic flux through the smaller loop is, to a good approximation, the product of this field and the area (πr^2) of the smaller loop:

$$\Phi_B = \frac{\pi \mu_0 i r^2 R^2}{2x^3}.$$

(b) The emf is given by Faraday's law:

$$\varepsilon = -\frac{d\Phi_B}{dt} = -\left(\frac{\pi \mu_0 i r^2 R^2}{2}\right) \frac{d}{dt} \left(\frac{1}{x^3}\right) = -\left(\frac{\pi \mu_0 i r^2 R^2}{2}\right) \left(-\frac{3}{x^4} \frac{dx}{dt}\right) = \frac{3\pi \mu_0 i r^2 R^2 v}{2x^4}.$$

(c) As the smaller loop moves upward, the flux through it decreases. The induced current will be directed so as to produce a magnetic field that is upward through the smaller loop, in the same direction as the field of the larger loop. It will be **counterclockwise** as viewed from above, in the same direction as the current in the larger loop.

11-2

(a) We assume the flux is entirely due to the field generated by the long straight wire. We find the absolute value of the flux.

$$|\Phi_B| = \int_{r-b/2}^{r+b/2} \left(\frac{\mu_0 i}{2\pi r}\right) (a dr) = \frac{\mu_0 i a}{2\pi} \ln \left(\frac{r+b/2}{r-b/2}\right).$$

(b) Implementing Faraday's law involves taking a derivative of the flux in part (a), and recognizing that $dr/dt = v$. The magnitude of the induced emf divided by the loop resistance then gives the induced current:

$$i_{\text{loop}} = \left| \frac{\varepsilon}{R} \right| = -\frac{\mu_0 i a}{2\pi R} \left| \frac{d}{dt} \ln \left(\frac{r+b/2}{r-b/2}\right) \right| = \frac{\mu_0 i a b v}{2\pi R [r^2 - (b/2)^2]}$$

11-3

(a) The coil-solenoid mutual inductance is

$$M = M_{cs} = \frac{N\Phi_{cs}}{i_s} = \frac{N(\mu_0 i_s n \pi R^2)}{i_s} = \mu_0 \pi R^2 n N .$$

(b) As long as the magnetic field of the solenoid is entirely contained within the cross-section of the coil we have $\Phi_{sc} = B_s A_s = B_s \pi R^2$, regardless of the shape, size, or possible lack of close-packing of the coil.

11-4

In absolute value, Faraday's law (for a single turn, with B changing in time) gives

$$\frac{d\Phi_B}{dt} = \frac{d(BA)}{dt} = A \frac{dB}{dt} = \pi R^2 \frac{dB}{dt}$$

for the magnitude of the induced emf. Dividing it by R^2 then allows us to relate this to the slope of the graph in Fig. (b) [particularly the first part of the graph], which we estimate to be $80 \mu\text{V}/\text{m}^2$.

(a) Thus, $\frac{dB_1}{dt} = (80 \mu\text{V}/\text{m}^2)/\pi \approx 25 \mu\text{T}/\text{s} .$

(b) Similar reasoning for region 2 (corresponding to the slope of the second part of the graph in Fig. (b)) leads to an emf equal to

$$\pi r_1^2 \left(\frac{dB_1}{dt} - \frac{dB_2}{dt} \right) + \pi R^2 \frac{dB_2}{dt} .$$

which means the second slope (which we estimate to be $40 \mu\text{V}/\text{m}^2$) is equal to $\pi \frac{dB_2}{dt}$. Therefore, $\frac{dB_2}{dt} = (40 \mu\text{V}/\text{m}^2)/\pi \approx 13 \mu\text{T}/\text{s}.$

(c) Considerations of Lenz's law leads to the conclusion that B_2 is increasing.

11-5

The induced electric field E as a function of r is given by $E(r) = (r/2)(dB/dt)$.

(a) The acceleration of the electron released at point a is

$$\vec{a}_a = \frac{eE}{m} \vec{i} = \frac{er}{2m} \left(\frac{dB}{dt} \right) \vec{i} = \frac{(1.60 \times 10^{-19} \text{ C})(5.0 \times 10^{-2} \text{ m})(10 \times 10^{-3} \text{ T/s})}{2(9.11 \times 10^{-31} \text{ kg})} \vec{i} = (4.4 \times 10^7 \text{ m/s}^2) \vec{i}.$$

(b) At point b we have $a_b \propto r_b = 0$.

(c) The acceleration of the electron released at point c is

$$\vec{a}_c = -\vec{a}_a = -(4.4 \times 10^7 \text{ m/s}^2) \hat{i}.$$

11-6

The flux Φ_B over the toroid cross-section is

$$\Phi_B = \int_a^b B dA = \int_a^b \left(\frac{\mu_0 N i}{2\pi r} \right) h dr = \frac{\mu_0 N i h}{2\pi} \ln \left(\frac{b}{a} \right).$$

Thus, the coil-toroid mutual inductance is

$$M_{ct} = \frac{N_c \Phi_{ct}}{i_t} = \frac{N_c}{i_t} \frac{\mu_0 i_t N_t h}{2\pi} \ln \left(\frac{b}{a} \right) = \frac{\mu_0 N_1 N_2 h}{2\pi} \ln \left(\frac{b}{a} \right)$$

where $N_t = N_1$ and $N_c = N_2$.

12-1 . (a) At any instant the displacement current I_d in the gap between the plates equals the conduction current I in the wires. Thus $I_d = I = 2.0 \text{ A}$.

(b) The rate of change of the electric field is

$$\frac{dE}{dt} = \frac{1}{A} \left(\frac{d\Phi_E}{dt} \right) = \frac{1}{\epsilon_0 A} \left(\epsilon_0 \frac{d\Phi_E}{dt} \right) = \frac{I_d}{\epsilon_0 A} = \frac{2.0 \text{ A}}{(8.85 \times 10^{-12} \text{ F/m})(1.0 \text{ m})^2} = 2.3 \times 10^{11} \text{ V} \cdot \text{m/s}$$

(c) The displacement current through the indicated path is

$$I'_d = I_d \left(\frac{d^2}{L^2} \right) = (2.0 \text{ A}) \left(\frac{0.50 \text{ m}}{1.0 \text{ m}} \right)^2 = 0.50 \text{ A}$$

(d) The integral of the field around the indicated path is

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I'_d = (4\pi \times 10^{-7} \text{ H/m})(0.50 \text{ A}) = 6.3 \times 10^{-7} \text{ T} \cdot \text{m}$$

12-2. (a) The current in the wires is the rate at which charge is accumulating on the plates and must also be the displacement current in the capacitor:

$$I = I_{d,\max} = 35\mu\text{A}$$

(b) At any instant, the charge on the plates is

$$Q = C \varepsilon (emf)$$

So the current is $I_d = \frac{dQ}{dt} = -\omega\varepsilon_0(emf) \sin \omega t (\varepsilon_0 \frac{A}{d})$, or

$$I_{d,\max} = \omega\varepsilon_0(emf)(\varepsilon_0 \frac{A}{d})$$

$35 \times 10^{-6} \text{ A} = 2\pi(96.0 \text{ Hz})\varepsilon_0(emf)(8.85 \times 10^{-12} \text{ F/m})\pi(0.025 \text{ m})^2 / (2.0 \times 10^{-3} \text{ m})$,which gives

$$\varepsilon_0(emf) = 6.68 \times 10^3 \text{ V}$$

(c) We can find the maximum value of from the maximum value of the displacement current

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

$$(\frac{d\Phi_E}{dt})_{\max} = \frac{I_{d,\max}}{\varepsilon_0} = 3.95 \times 10^6 \text{ V} \cdot \text{m/s}$$